Not-ruled-out images of policy contrasts

A village agent-based model, calibrated by history matching, and the analysts it can grade

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ABSTRACT

A calibrated simulator answers a policy question with one number, yet many parameter points can fit its targets about equally well. We study a planted village, an agent-based model of preventive behavior on a fixed network, in which a planner asks what seeding the best-connected people would do against no campaign. The targets are what a researcher could observe: the status quo and one randomized pilot. History matching keeps the parameter points that the targets cannot rule out, and we report the image of the policy effect over them. Because the truth is known, we grade one-panel intervals in two separate ways: whether they contain the truth, and whether they are narrower than what the targets leave open. In the one panel at the real village, the analyst who assumes no interference reports a tight interval that misses the policy effect. At the calibrator's two best-fitting points, the same analyst's interval misses again and is overconfident: it lies strictly inside the range of effects that the targets leave open. Over repeated panels, analysts who allow for interference through an exposure map and transport from the pilot reduce the absolute bias at every point, at the price of a larger sampling spread and intervals that are conservative relative to it; in the one panel at the calibrator's points, their intervals contain the effect at her point and still lie strictly inside that range. A calibrated simulator should report the image, and a one-point interval should be read against it. The exercise is a planted methods illustration and makes no claim about a real campaign.

KEYWORDS partial identification; history matching; interference; exposure mappings; agent-based models; calibration

JEL CODES C15, C18, C31, C63, D85

Code and replication: https://github.com/Bonorinoa/ecn593-applied-project (private until the author's review). Design: `docs/preregistration-v2.md`, with two amendments written after earlier results (Appendix G).

SECTION 1

Introduction

A planner wants to know what a campaign would do to preventive behavior in a village if it seeded the 80 best-connected people, compared with no campaign at all. Seeding the people with the most contacts is a common proposal for network interventions, and field work has studied both its promise and its limits (Banerjee et al., 2013; Kim et al., 2015). Before the campaign, the planner can see the village as it is, and she may be able to run one randomized pilot. She cannot see the policy itself. A common way to answer the question is to build a simulator of the village, calibrate it to observable moments, and run the policy inside the simulator (Kydland and Prescott, 1996). The answer is then one number: the policy effect at the calibrated parameter point, perhaps with a Monte Carlo error bar.

That number can mislead for a reason that has little to do with the simulator's quality. Calibration chooses one parameter point, but the targets rarely pin one down. Many points can match the same moments about equally well, a fact that simulation modellers call equifinality (Beven, 2006) and that economists have long raised about calibration (Hansen and Heckman, 1996). If those points imply different policy effects, the single number carries an extra, unstated assumption: that the chosen point is the world. History matching takes the opposite stance. It rules out the parameter points that the targets reject and keeps the rest, the not-ruled-out-yet (NROY) set (Craig et al., 1997; Vernon et al., 2010; Andrianakis et al., 2015). The natural report is then the *image* of the policy effect over that set. This is the reporting logic of partial identification (Manski, 2003; Tamer, 2010): when the maintained assumptions leave a set of answers, report the set rather than a convenient point.

The policy question has a second difficulty. People in a village influence each other, so the outcome of one person depends on who else is seeded. The policy effect is then a contrast of two whole assignment vectors, the one that seeds the hubs and the one that seeds no one (Halloran and Struchiner, 1995; Hudgens and Halloran, 2008; Manski, 2013). A randomized pilot seeds people at random. It realizes the hub-seeding vector only with negligible probability, so an estimate of the policy effect from the pilot needs two further assumptions: an exposure map, under which a person's outcome depends on the assignment only through a low-dimensional summary, and transport, under which the outcomes measured on the pilot's people apply to the people who would hold the same exposures under the policy. An analyst who ignores interference avoids both assumptions only by making a stronger one.

This paper puts the two difficulties side by side in a setting where the truth is known. The village is an agent-based model with 800 people on a fixed random graph. The targets are planted: they are the model's own moments at a chosen point $c^\star$, the real village of the exercise. They are what a researcher could observe, the status quo and one randomized pilot; the policy to predict is never a target. History matching keeps the points that the targets cannot rule out, and we report the image of the policy effect over a declared sample of them. We then hand one panel to several analysts, each with an interval for the policy effect (or, in one case, for a spillover contrast), and we grade each interval in two separate ways: whether it contains the truth, and how it sits against the image. The analysts form a ladder of assumptions. The first assumes no interference. Two more allow for interference through an exposure map and transport from the pilot, the second of them without borrowing control outcomes across

a split of the village into hubs and everyone else. They work for two kinds of researcher: a field experimenter who has one panel from the real village, and a calibrator who works at one best-fitting point of the model.

The contribution can be stated in three sentences. First, we write the policy effect under interference as a contrast of two assignment vectors and make explicit what an estimate of it from a randomized pilot must assume. Second, we report the image of that effect over the parameter points that observable targets do not rule out, and we grade one-panel intervals against both the truth and the image, keeping containment and interval geometry apart. Third, we measure over repeated panels what an analyst gains and loses by allowing for interference: less bias, more spread, and intervals that are conservative relative to their sampling distribution.

> **PROPOSED HEADLINE (OWNER DECISION)** In the one panel at the real village, the analyst who assumes no interference reports a tight interval that misses the policy effect. At the calibrator's two best-fitting points, the same analyst's interval misses again and is overconfident: it lies strictly inside the range of effects that the targets leave open. Over repeated panels, analysts who allow for interference through an exposure map and transport from the pilot reduce the absolute bias at every point, at the price of a larger sampling spread and intervals that are conservative relative to it; in the one panel at the calibrator's points, their intervals contain the effect at her point and still lie strictly inside that range. Report the image, and read any one-point interval against it.

Two words in the headline carry precise meanings, and we keep them apart throughout. An interval *misses* when it does not contain the true value at the point in question; this is a statement about one realized panel and one world. An interval is *overconfident*, in the sense of the verdict F-OVERCONF (Section 6.1), when it lies strictly inside the image: it is narrower than the set of effects that the targets leave open. An interval can contain the truth and still be overconfident, and it can miss the truth without being overconfident. We apply the charge of overconfidence only to the calibrator, who adds the assumption that her point is the world. The field experimenter works with the real village's panel; we report how her interval sits against the image, but that comparison is not a test of her estimator.

None of the pieces is new. History matching, partial identification, exposure mappings and robust standard errors are all standard, and the paper treats each of them as a well-built tool for the question it was designed to answer. What we add is their composition around one named policy contrast, in a planted setting that lets every interval be checked. The numbers are the simulator's own, so the paper is a methods illustration with no claim about any real campaign.

Section 2 defines the estimand and states what the pilot can and cannot reveal about it. Section 3 describes the village and the targets, and Section 4 describes history matching and the image. Section 5 sets out the ladder of analysts, and Section 6 the evaluation. Section 7 reports the results, Section 8 relates the procedure to partial identification, Section 9 lists what the procedure is not, and Section 10 concludes. The appendix holds the model details, the checks behind each analyst, the sensitivity menu, replication and the disclosures.

SECTION 2

The estimand

2.1 Policies are assignment vectors

The village has $N = 800$ people, a fixed undirected graph with adjacency matrix A, and covariates X. An assignment is a vector $\mathbf{d} \in \{0, 1\}^N$, where $d_i = 1$ means that person i is seeded. A policy is a rule that produces an assignment. We use three:

- π_0, the status quo, seeds no one: $\mathbf{d} = \mathbf{0}$.
- π_1, the pilot, seeds $K = 80$ people drawn uniformly at random without replacement, so $K/N = 0.1$.
- π_2, the policy to predict, seeds the K people with the highest degree, with ties broken toward the lower index. Given the graph it is a single fixed vector, which we write $\mathbf{d}_2$.

A parameter point c indexes a complete data-generating process: for each c, each draw ω of the simulation noise, and each assignment $\mathbf{d}$, the kernel returns the outcomes of everyone at the horizon T (Section 3). Under interference, the outcome of person i depends on the whole vector, $Y_{iT} = Y_{iT}(\mathbf{d})$, so the effect of seeding person i alone is not defined until the assignment of everyone else is named. The well-defined object is a contrast of policies,

$$\theta(\pi, \pi'; c) \;=\; \mathbb{E}_c\left[\bar{Y}_T(\pi)\right] \;-\; \mathbb{E}_c\left[\bar{Y}_T(\pi')\right], \tag{1}$$

where $\bar{Y}_T$ is the mean outcome over people and the expectation is over the simulation noise and, for π_1, over the random assignment. This is an overall effect of one assignment rule against another in the sense of Halloran and Struchiner (1995) and Hudgens and Halloran (2008). The code defines three such contrasts:

$$\theta_1(c) = \theta(\pi_1, \pi_0; c), \tag{2}$$

$$\theta_2(c) = \theta(\pi_2, \pi_0; c) = \mathbb{E}_c\left[\bar{Y}_T(\mathbf{d}_2)\right] - \mathbb{E}_c\left[\bar{Y}_T(\mathbf{0})\right], \tag{3}$$

$$\psi(c) = \theta_2(c) - \theta_1(c). \tag{4}$$

The headline estimand is θ_2, the effect of seeding the hubs against no campaign, and ψ is the premium of targeting over random seeding. In the code each expectation is a mean over 200 paired replications: rep r of every arm shares one state stream (the starting state and every shock), so the contrasts between arms carry little Monte Carlo noise. The estimands are defined on the terminal mean $\bar{Y}_T$.

Equation (3) makes the main point of this section. The headline estimand compares two specific vectors, $\mathbf{d}_2$ and $\mathbf{0}$. It is a property of those two vectors, of the graph and of the kernel at c. Any data that never contain $\mathbf{d}_2$ can speak to it only through assumptions that connect what they do contain to what $\mathbf{d}_2$ would produce.

2.2 What a randomized pilot reveals

A pilot draws one assignment from π_1. Under that design every vector with exactly K ones has the same probability, so the pilot realizes the policy vector with probability

$$\Pr_{\pi_1}(\mathbf{d} = \mathbf{d}_2) = \binom{N}{K}^{-1} = \binom{800}{80}^{-1}. \tag{5}$$

That probability is positive and negligible. The pilot realizes the policy with negligible probability, so in practice an estimate of θ_2 from it needs an exposure map and transport.

What the pilot does reveal is θ_1, the effect of the pilot's own design. The difference between the mean outcome in the pilot and the mean outcome in the status quo estimates θ_1 without further assumptions. In the planted exercise this holds exactly by construction: the pilot and status-quo targets share their state streams, so their difference at $c^\star$ equals $\theta_1(c^\star)$ (Section 3.2). The question is whether, and under which assumptions, one can go from the pilot to θ_2.

2.3 What an estimate from the pilot must assume

The standard route goes through exposures (Aronow and Samii, 2017; Leung, 2020). Let $k_i = \max(\sum_j A_{ij}, 1)$ and let $s_i(\mathbf{d}) = \sum_j A_{ij} d_j / k_i$ be the share of i's neighbors who are seeded. The code uses four exposure strata,

$$S_1 = \{i : d_i = 1\}, \qquad S_{0a} = \{i : d_i = 0,\ s_i = 0\},$$
$$S_{0b} = \{i : d_i = 0,\ 0 < s_i < 0.25\}, \qquad S_{0c} = \{i : d_i = 0,\ s_i \geq 0.25\},$$

and we write $\mathcal{S} = \{S_1, S_{0a}, S_{0b}, S_{0c}\}$ and $e_i(\mathbf{d}) \in \mathcal{S}$ for the stratum of person i under $\mathbf{d}$. Two assumptions take the pilot to θ_2.

Assumption 1 (Exposure map). The expected outcome of each person depends on the assignment only through the person's stratum: $\mathbb{E}_c[Y_{iT}(\mathbf{d})] = \mathbb{E}_c[Y_{iT}(\mathbf{d}')]$ whenever $e_i(\mathbf{d}) = e_i(\mathbf{d}')$.

Let μ_S be the exposure-S mean outcome that the pilot measures: the inverse-probability-weighted (Hájek) mean over the people who land in S, each weighted by the inverse of the probability that the pilot's lottery puts them there (Horvitz and Thompson, 1952; Aronow and Samii, 2017).

Assumption 2 (Transport). For each stratum S, the exposure-specific outcome μ_S measured on the pilot's people applies to the people who would hold exposure S under $\mathbf{d}_2$.

Under the status quo nobody is seeded, so everyone is in S_{0a}. Every person can land in S_{0a} under the pilot, so under Assumption 1 the mean $\mu_{S_{0a}}$ is the status-quo mean of the whole village. Under the policy, a known share $w_S = n_S(\mathbf{d}_2)/N$ of the village holds exposure S; the analyst knows the graph, so the shares are known constants that sum to one. Under both assumptions,

$$\theta_2(c) \;=\; \sum_{S \in \mathcal{S}} w_S\, \mu_S - \mu_{S_{0a}} \;=\; \sum_{S \neq S_{0a}} w_S \left(\mu_S - \mu_{S_{0a}}\right). \tag{6}$$

Every term on the right can be estimated from the pilot, because the pilot's lottery is known and the probability that each person lands in each stratum can be computed before any outcome is seen. Equation (6) is the target of the analyst A1 in Section 5.

Neither assumption holds exactly in this village, and the paper does not pretend that it does. The kernel's peer term uses neighbors' lagged outcomes, so the effect of a seed travels beyond its own neighbors; a person in S_{0a} in the pilot can still be lifted by a seed two steps away. The exposure map therefore misses part of the interference. Transport fails in a second way: under $\mathbf{d}_2$ the seeded people are the hubs, while under the pilot they are a random draw, and hubs need not respond like a random draw. The analyst A2 relaxes this second failure by splitting the village into the hubs and everyone else and transporting only within each part; for the hubs, its term compares the hubs' own seeded and unexposed outcomes in the pilot. Neither analyst identifies θ_2 from the pilot. Without assumptions on how outcomes respond to the assignments of others, the data from one randomized design bound the effects of other assignment rules only loosely (Manski, 2013). The analysts below differ in which assumptions they add, and Section 7 measures what each assumption costs and buys.

The analyst who assumes no interference at all (A0 below) avoids the exposure map and transport, but only by assuming that a person's outcome depends on her own seeding alone (Rubin, 1980). Under that assumption, θ_2 equals the seeded share times the effect of own seeding. Under interference it does not, and the gap between the two is measurable here (Appendix B).

SECTION 3

The village model and the targets

3.1 The village

The world is a graph and a matrix of covariates, drawn once with fixed seeds and shared by every simulation. The graph follows an edge rule rather than a target mean degree: for every ordered pair the code draws $A_{ij} \sim \text{Bernoulli}(p)$ with $p = 4/N$, symmetrizes the result and sets the diagonal to zero. The expected mean degree is 7.97, and the drawn graph has mean degree 8.085. It is close to tree-like (Appendix A). Each person has three covariates $X_i \sim U[0,1]^3$, which stand for education, resources and a baseline propensity.

Each person has a level of preventive behavior $Y_{it} \in (0,1)$. Starting from $Y_i(0) \sim U[0,1]$, the kernel updates every person at once for $t = 0, \ldots, T-1$ with $T = 20$:

$$Y_i(t+1) \;=\; \Lambda\Big(\alpha + \beta\, Y_i(t) + \gamma\, \frac{(A\,Y(t))_i}{k_i} + \eta^\top X_i + \kappa\, d_i + \sigma_\varepsilon\, \varepsilon_i(t)\Big), \tag{7}$$

where $\Lambda(u) = 1/(1+e^{-u})$, $\varepsilon_i(t) \sim N(0,1)$ and seeding acts from the first period on. The index is linear in means, passed through a logistic link. The update rule is chosen, not derived; its lineage is the linear-in-means model of Manski (1993), the row-normalized peer mean on a network of Bramoullé et al. (2009), and the logistic link of discrete choice with social interactions in Brock and Durlauf (2001). Peers enter through their lagged outcomes, not through their seeding status, so the effect of a

Table 1. The free parameters, their box C, and the planted point $c^\star$.

Parameter	Box	Planted value	Role
α	$[-2.5, -0.5]$	-1.5	intercept of the index
β	$[0.05, 0.60]$	0.325	weight on the own lag
γ	$[0.05, 0.80]$	0.425	weight on the peer mean
κ	$[0.25, 2.50]$	1.375	effect of being seeded
σ_ε	$[0.01, 0.40]$	0.205	SD of the shock
η	fixed	$(0.05, 0.10, 0.02)$	covariate weights

The planted point is the center of C. The box, η, the graph rule, N, K and T are exogenous and live in `config/world.toml`.

Table 2. The target moments, in the order the code uses.

#	Name	Arm	Target z	Definition
1	`sq_level`	status quo	0.2255	mean over people of Y_T
2	`sq_corr_x1`	status quo	0.0726	correlation of Y_T with the first covariate
3	`sq_peer_corr`	status quo	0.0003	correlation of Y_{iT} with the peer mean $(AY_T)_i/k_i$
4	`pilot_level`	pilot	0.2618	mean over people of Y_T

Targets are planted: $z = \widehat{M}(c^\star)$ with 200 replications, stored in `data/planted.json`.

seed spreads along the graph over time. This is the source of the interference that the analysts must handle.

Five parameters are free, $c = (\alpha, \beta, \gamma, \kappa, \sigma_\varepsilon)$, and they range over a box C fixed before any run (Table 1). The covariate weights η are fixed inside the kernel. The box excludes $\gamma = 0$ on purpose, since the project is about interference. Over the whole box the update map is a contraction in the sup norm, so for a fixed assignment the process has a unique stationary regime and forgets its starting state at a geometric rate (Appendix A). The kernel has no tipping points and no multiple equilibria; in the language of Section 8, it is complete.

3.2 The targets: the status quo and one randomized pilot

The targets are what a researcher could observe. She sees the village as it is, under π_0, and she can run one randomized pilot, under π_1. She never sees the policy π_2. That policy is what she wants to predict, so it is never a target; if it were, the targets would contain the answer to the planner's question.[1]

The code uses 4 moments in a fixed order (Table 2). Each is a mean over 30 paired replications, and rep r of the status-quo arm and of the pilot arm share one state stream. The Monte Carlo variance of each moment is the sample variance over the replications divided by the number of finite values; the code drops non-finite correlations, and if fewer than two values are finite the variance is infinite and the point is rejected.

[1] An earlier design of this project did exactly that: its targets used outcomes under both seeding policies, and their difference was the targeting premium. It is archived at the git tag `v1-worldbank-moments` and plays no role here.

The targets are planted: $z = \widehat{M}(c^\star)$, the simulated moments at the planted point, computed with 200 replications together with their Monte Carlo variances. Planting gives a known benchmark. It does not guarantee that a finite search keeps any point, and it makes the true effect known, which is the only way to check an interval against the truth.

Because the status-quo and pilot arms share their state streams, the difference `pilot_level − sq_level` at $c^\star$ equals $\theta_1(c^\star)$ exactly: the two printed values are 0.03631 and 0.03631. This is intended. A randomized pilot identifies the effect of its own design. The question of the paper is what it says about θ_2, a contrast that no target contains.

SECTION 4

History matching and the image

4.1 Ruling points out

History matching asks which parameter points the targets cannot rule out. Let $\widehat{M}(c)$ be the simulated moments at c. The implausibility of c is

$$I(c) \;=\; \big(z - \widehat{M}(c)\big)^{\top} V(c)^{-1} \big(z - \widehat{M}(c)\big), \qquad V(c) = \operatorname{diag}\big(\sigma_z^2 + \sigma_{\mathrm{MC}}^2(c) + \delta^{\circ 2}\big), \tag{8}$$

a variance-weighted sum of squared gaps between targets and simulated moments. The three terms of V are the Monte Carlo variance of the targets, the Monte Carlo variance of the candidate's moments, and a fixed discrepancy δ, the allowance for a model that does not match the world exactly (Kennedy and O'Hagan, 2001; Vernon et al., 2010). In this planted exercise δ is a design tolerance rather than an estimated model-error variance: $\delta = (0.02,\ 0.05,\ 0.05,\ 0.02)$ in the order of Table 2, that is, 0.02 for the two levels and 0.05 for the two correlations. A point is kept if $I(c) \le \tau$, where $\tau = 9.4877$ is the 0.95 quantile of a χ^2 distribution with 4 degrees of freedom.

The threshold is a rule for ruling out. It is not the critical value of a test, and it does not define a credible region: the two level moments share a state stream, V is diagonal, and δ^2 rather than sampling noise fills most of V. Moving τ or δ after seeing which points survive would turn a filter into a fit, so both are fixed before any run. Appendix E reports, as a disclosed sensitivity analysis, how the image changes when the level tolerance is tightened.

The search uses no emulator; every point is simulated. Wave 1 is a Latin hypercube of 1000 points in C. Waves 2 and 3 each draw 1000 new points in the bounding box of the points kept by the previous wave, widened by 20% of its span and clipped to C. In the code, NROY is a finite set: the union over the three waves of the sampled points with $I(c) \le \tau$. It approximates the level set $\{c \in C : I(c) \le \tau\}$ without enumerating it. Membership means that the targets cannot rule a point out. It does not mean the point is true, and NROY is not a posterior over c. An empty NROY would be a finding: the pair of kernel and targets would be rejected.

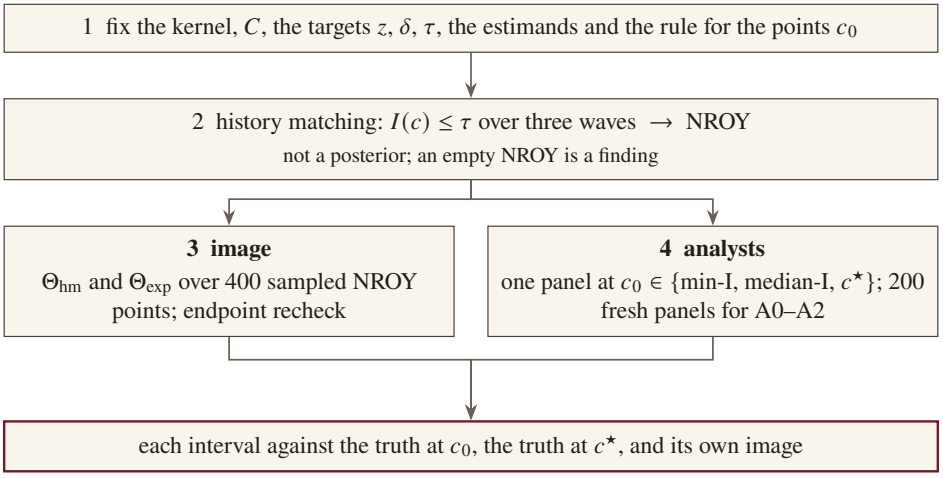


Figure 1. Order of operations. The image and the analysts are two separate reports; the evaluation compares them and never merges them into one object.

4.2 The image

The object we report is the image of the policy effect over the kept points. Let $\mathcal{S}$ be a declared uniform sample of 400 NROY points. The code reports

$$\Theta_{hm} = \Big[\min_{c\in\mathcal{S}} \theta_2(c),\ \max_{c\in\mathcal{S}} \theta_2(c)\Big], \qquad \Theta_{exp} = \Big[\min_{c\in\mathcal{S}} \delta_{spill}(c),\ \max_{c\in\mathcal{S}} \delta_{spill}(c)\Big], \tag{9}$$

where δ_{spill} is the spillover contrast defined with Foil 2 in Section 5.2. Each image is the range over the sample, the declared empirical image. It is a set, not a distribution: the range says which values the targets leave open and says nothing about how likely each one is.

The endpoints are extrema of noisy estimates, so the code evaluates each endpoint candidate again on fresh replications (Appendix D).

4.3 The diagnostic points

The analysts of Section 5 work at three diagnostic points, which we write c_0. Two belong to the calibrator, who fits the model and works at one best-fitting point: the *min-I point*, the NROY point with the smallest implausibility, and the *median-I point*, the point at the middle of NROY in the order of I. Both are selected by I alone, never by θ_2. The third is $c^\star$ itself, the real village, which belongs to the field experimenter. We use three points rather than all 400: running an analyst at every kept point would hand her the calibration set, and default practice does not do that.

Figure 1 summarizes the order of operations. Every exogenous quantity is fixed before any wave runs. If θ_2 decided which points to keep, the image would be a fit rather than a report.

Table 3. The analysts. A0, A1 and A2 aim at θ_2; Foil 2 aims at the spillover δ_{spill} and is compared only with its own image.

Analyst	Panel	Target	Maintained assumptions
A0 (Foil 1)	policy arm	θ_2	no interference; unconfoundedness given X; linear constant-effect model
A1	pilot	θ_2	exposure map; transport from the pilot with the policy's exposure shares
A2	pilot	θ_2	exposure map; transport within the hubs and within everyone else
Foil 2	pilot	δ_{spill}	exposure map; its target is a spillover contrast

A0 is compared with Θ_{hm}, as are A1 and A2; Foil 2 is compared with Θ_{exp}. A0 and Foil 2 are the two foils of the main pre-registration; A1 and A2 were added by its amendment A2.

SECTION 5

The analysts: a ladder of assumptions

Each analyst sees one panel at a diagnostic point c_0: the outcomes Y_T, the assignment, the covariates and the graph, from one draw of the simulation noise. None of them sees NROY. Three analysts report an interval for θ_2; one, Foil 2, reports an interval for a different contrast, the spillover on the pilot. Every interval has nominal level 95% and uses the two-sided normal quantile. Table 3 lists them in order of decreasing assumption strength. A0, A1 and A2 are three intervals aimed at θ_2 under different assumptions. They do not estimate the same functional, and only A0's needs SUTVA to equal θ_2.

5.1 A0: the analyst who assumes no interference (Foil 1)

A0 has one panel from the policy itself, the arm that seeds the hubs. She fits OLS of Y_T on $Z_i = (1, d_i, X_i^\top)$ and reads the coefficient on d_i, $\hat{\beta}$, with the heteroskedasticity-robust HC1 standard error (MacKinnon and White, 1985),

$$\widehat{\text{Var}}_{\text{HC1}}(\hat{\beta}) = \left[\tfrac{n}{n-p}\,(Z^\top Z)^{-1}\Big(\textstyle\sum_i \hat{u}_i^2\, Z_i Z_i^\top\Big)(Z^\top Z)^{-1}\right]_{dd}, \tag{10}$$

where p is the number of columns of Z. Under no interference, the policy effect for the whole village is the seeded share times the effect of own seeding, so she reports $(K/N)\hat{\beta}$ with the interval scaled the same way, and we call that interval J_1. The scaling by a known positive constant keeps her own coverage statement and does not repair her assumptions.

Her interval answers the question it was built for under three conditions: no interference, so that a person's outcome depends only on her own seeding; unconfoundedness given X; and a linear, constant-effect outcome model. The village breaks the first by construction. The second is not supported either: on the policy arm, seeding is a deterministic function of degree, which the regression does not include. A0 is still the natural default, and it is a good estimator of what it estimates. Under interference, $(K/N)\hat{\beta}$ measures something close to the effect of being one of the hubs on the hubs,

and it has no term for the effect of the campaign on everyone else. Appendix B decomposes the gap exactly and explains why a flexible regression on the same panel would not help.

5.2 Foil 2: the spillover on the pilot

Foil 2 uses the pilot panel and the exposure strata of Section 2.3. Because the pilot's lottery is known, the probability $p_{i,S}$ that person i lands in stratum S can be computed before any outcome is seen. For an unseeded person of degree k, the number of seeded neighbors under the pilot is hypergeometric, so the first-order probabilities are exact. Foil 2 replaces each stratum mean with a Hájek mean,

$$\widehat{\mu}_S = \frac{\sum_i \mathbf{1}\{i \in S\}\, Y_{iT}/p_{i,S}}{\sum_i \mathbf{1}\{i \in S\}/p_{i,S}}, \tag{11}$$

and estimates the spillover contrast $\widehat{\mu}_{S_{0c}} - \widehat{\mu}_{S_{0a}}$: the outcome of unseeded people with many seeded neighbors against unseeded people with none. Its target is

$$\delta_{\text{spill}}(c) = \overline{Y}_{S_{0c}} - \overline{Y}_{S_{0a}}, \tag{12}$$

the difference of the realized stratum means of the true outcomes, averaged over 200 pilot replications. The variance is the linearized estimator of Aronow and Samii (2017). Its joint exposure probabilities come from 10,000 simulated pilot assignments, and Young's bound, as used by Aronow and Samii (2017), covers pairs of people who never appear together in a pair of strata. If any stratum in a panel holds fewer than 20 people, the code merges S_{0b} into S_{0c} for that panel. We call the interval J_2.

Foil 2 is the analyst who takes interference seriously but answers a different question. Its target is δ_{spill}, not θ_2, and its interval is compared only with Θ_{exp}, the image of δ_{spill}. The Hájek mean targets an inverse-probability-weighted exposure contrast, while δ_{spill} averages the realized contrasts without weights; on this run the two are close enough that no containment flag depends on the difference (Appendix C). Foil 2 is the only analyst with a pre-registered calibration check of its interval, also in Appendix C.

5.3 A1: exposure transport

A1 uses the same pilot panel, the same strata, the same cut and the same merge rule as Foil 2, and the same Hájek means with exact first-order probabilities. It estimates Equation (6),

$$\widehat{\theta}_2^{\text{A1}} = \sum_{S \neq S_{0a}} w_S\big(\widehat{\mu}_S - \widehat{\mu}_{S_{0a}}\big), \qquad w_S = \frac{n_S(\mathbf{d}_2)}{N}, \tag{13}$$

where $n_S(\mathbf{d}_2)$ is the number of people in stratum S under the policy vector. When a panel is merged, the shares use the merged strata. The estimate is a fixed linear combination of Hájek means with coefficients a_S, so its variance is the same linearized covariance summed over all pairs of strata, $\sum_S \sum_{S'} a_S a_{S'} \widehat{\text{Cov}}(\widehat{\mu}_S, \widehat{\mu}_{S'})$, with joint probabilities from the same simulated assignments.

A1 maintains the exposure map and transport. It allows interference, but only the part that runs through the share of seeded neighbors, and it applies the pilot's exposure-specific outcomes to the people who would hold those exposures under the policy.

5.4 A2: degree-binned transport

A2 splits the village into two bins fixed by the graph: H, the K people whom $\mathbf{d}_2$ seeds, and L, everyone else. Under $\mathbf{d}_2$ every person in H is in S_1, so

$$\widehat{\theta}_2^{\mathrm{A2}} = \frac{K}{N}\left(\widehat{\mu}_{S_1,H} - \widehat{\mu}_{S_{0a},H}\right) + \sum_{S \neq S_{0a}} \frac{n_{S,L}(\mathbf{d}_2)}{N}\left(\widehat{\mu}_{S,L} - \widehat{\mu}_{S_{0a},L}\right), \tag{14}$$

where $\widehat{\mu}_{S,B}$ is the Hájek mean of stratum S within bin B. The variance uses the same machinery, with each cell's indicators restricted to its bin. A2 never merges across bins and never borrows a control mean from the other bin. If a cell that Equation (14) needs is empty in a panel, A2 is undefined for that panel; the code counts that case and never imputes it.

A2 is the weakest rung of the ladder. The hub term compares the hubs with themselves, so it needs the exposure map but no transport across people. The remaining terms transport only within L. The price is precision: A2 restricts each cell to one bin, so the cells it needs are thinner than those of A1, which pools them.

5.5 What the ladder predicts, written in advance

Amendment A2 of the pre-registration wrote five expectations before any A2 number existed. At every diagnostic point: (a) the absolute bias falls from A0 to A1 to A2; (b) the sampling SD and the interval width rise from A0 to A1 to A2; (c) A0's coverage is far below nominal, and A2's is the closest to nominal of the three; (d) A1 and A2 stay biased downward, because two-hop spillover raises the people in S_{0a} above the true status-quo level; (e) every interval is narrower than Θ_{hm}. Any of them could fail, and a failure is reported as found. Section 7.4 checks each one.

SECTION 6

Interval geometry and the evaluation

6.1 Two checks that answer different questions

Each interval is checked in two ways, and the two checks answer different questions. The first is *containment*: does the interval contain the true value of its target? The code records two flags. The flag `covers_truth` says whether the interval contains the truth at the point where the analyst works, $\theta_2(c_0)$ or $\delta_{\mathrm{spill}}(c_0)$. The flag `covers_planted_truth` says whether it contains the truth at the real village, $\theta_2(c^\star)$ or $\delta_{\mathrm{spill}}(c^\star)$. At $c^\star$ the two coincide. Both flags describe one realized panel. They are containment on one draw and are not estimates of repeated-sampling coverage.

The second check is *interval geometry*: how does the interval sit against the image of the same functional? Intervals and images are subsets of the real line, so a verdict is an inclusion. Let $J = [lo, hi]$ be an analyst's interval at one point, let $\Theta = [L, U]$ be the image of its own target, and write $\text{int}(\Theta) = (L, U)$. The function `classify_interval` returns

$$\text{F-HONEST} \iff lo \le L \text{ and } hi \ge U \iff \Theta \subseteq J, \tag{15}$$

$$\text{F-OVERCONF} \iff lo > L \text{ and } hi < U \iff J \subset \text{int}(\Theta), \tag{16}$$

and no verdict in every other case, including an interval that overlaps the image with neither inclusion. A0, A1 and A2 are compared with Θ_{hm}; Foil 2 is compared with Θ_{exp}.

F-OVERCONF means that the interval is narrower than the set of values that the targets leave open. It does not mean that the interval misses the truth. An interval can contain the truth and be F-OVERCONF at the same time: it can be right about its own world and still understate how much the targets leave open. Conversely, an interval can miss the truth without being F-OVERCONF. We therefore never use "overconfident" to mean "misses the truth," and we report the two checks side by side.

6.2 Two researchers, two layers

The three diagnostic points serve two researchers who know different things.

The field experimenter. She has one panel from the real village, the point $c^\star$, and she has calibrated nothing. Her question is whether her interval contains the truth there. We call this layer (a): is the estimator right about its own world? The answer depends only on the identification conditions of her estimator. The code also places her interval against the image, and we report that comparison, but it is not a test of her estimator: she never used the model, so a verdict against the image does not charge her with anything.

The calibrator. She fits the model to the targets and works at one best-fitting point, the min-I point or the median-I point. Her interval is checked three times. Layer (a) asks whether it contains the truth at her own point. Layer (b) asks whether it contains the truth at the real village; this adds the maintained assumption that her point is the world. The geometric verdict asks whether her interval is narrower than what the targets leave open. She is the analyst who replaces a set of not-ruled-out worlds with one of them, so the charge of overconfidence applies to her.

6.3 What the image maintains

The image is honest only relative to what it maintains: the kernel, the box C, the discrepancy δ and the targets with the threshold τ. Change any of them and the image changes. There is also a sampling caveat. The image is a range over 400 points drawn from a finite set of kept points, so it can understate the range over the full level set of I. A verdict of F-OVERCONF survives that error, because a larger image would still contain the interval strictly inside. A verdict of F-HONEST would not survive it.

6.4 Repeated panels for the three analysts of θ_2

The flags above describe one panel. For A0, A1 and A2, amendment A2 adds a repeated-sampling exercise at each diagnostic point: $R = 200$ fresh panels on a block of stream offsets that starts at 2000000 and is disjoint from every other block in the project. Each panel redraws the simulation noise and, for A1 and A2, the pilot's assignment. For each analyst the code reports the mean estimate, the bias (mean minus $\theta_2(c_0)$), the standard deviation of the estimates across panels (the sampling SD), the root mean squared error, the mean reported standard error and its ratio to the sampling SD, the share of panels whose interval contains $\theta_2(c_0)$ with an exact Clopper–Pearson 95% interval (Clopper and Pearson, 1934), the mean interval width, that width relative to the width of Θ_{hm}, and, for A2, the number of undefined panels.

These quantities measure different things, and we keep them apart. The sampling SD is the dispersion of the estimator. The mean reported SE is what the analyst believes that dispersion to be, and the mean width is the interval she reports. The repeated-sampling containment rate is the only coverage statement in this part of the paper, and the truth it uses is $\theta_2(c_0)$, the truth at the point that generated the panels. Addendum A2.1, written after these results were seen, splits A1's and A2's reported variance into its parts; it is descriptive and is reported with the results (Section 7.4).

SECTION 7

Results

Every number in this section is a macro generated from the committed results in `results/`. The section follows the order of the procedure: what history matching kept, the image, the one-panel intervals at the three points, the price of allowing for interference over repeated panels, and how the image moves with the level tolerance.

7.1 History matching kept a large set

Against the planted targets, the three waves kept 181, 449 and 408 of their 1000 points, and their union is $|\text{NROY}| = 1038$ (Figure A.1 in Appendix A). The set did not collapse to a point. The closest point has $I = 0.0141$, far below $\tau = 9.4877$, yet it does not recover $c^\star$: its parameters are $\alpha = -1.5956$, $\beta = 0.4709$, $\gamma = 0.6853$, $\kappa = 1.2496$ and $\sigma_\varepsilon = 0.2150$, against the planted values in Table 1. A good fit does not pick out the truth. The median-I point has $I = 3.3437$ and a larger seeding effect, $\kappa = 1.9202$.

Which parameters do the targets restrict? For each parameter, the code divides its standard deviation over the sampled NROY points by the standard deviation of a uniform draw over the box on that axis. A ratio near one means the targets do not restrict the parameter. The readout was declared before the run and is not a gate.

Only the intercept α is pinned down (Table 4). The ratios for β, γ and κ are close to one, and σ_ε is in between. The pre-registration expected the peer-correlation moment to give a ratio below one for γ. The ratio is 0.992, below one only in the third decimal, which is no restriction; that expectation

Table 4. Spread ratios over the sampled NROY points.

	α	β	γ	κ	σ_ε
spread ratio	0.259	0.965	0.992	0.962	0.839

Standard deviation over the 400 sampled NROY points divided by the standard deviation of a uniform draw over C on the same axis.

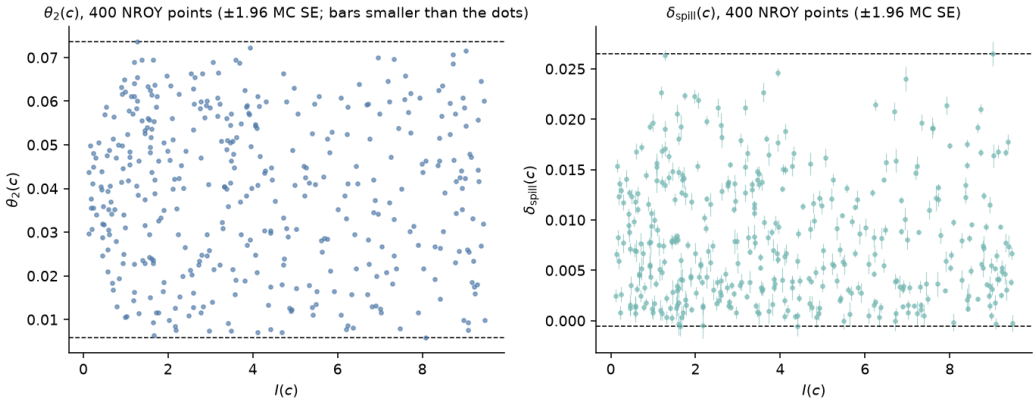


Figure 2. The 400 sampled NROY points against their implausibility $I(c)$. Left: $\theta_2(c)$. Right: $\delta_{\text{spill}}(c)$. Bars show ±1.96 pointwise Monte Carlo standard errors at each fixed c; dashed lines mark the image endpoints.

was not met. It also expected the pilot moment to restrict the trade-off between α and κ more than the earlier design did. This paper does not rerun the earlier design, so it cannot check that comparison; the sampled correlation between α and κ is -0.305.

7.2 The image of the policy effect is wide

At the planted point, $\theta_1(c^\star) = 0.03631$ and $\theta_2(c^\star) = 0.03826$, so the premium of targeting is $\psi(c^\star) = 0.00195$: in this village, seeding the hubs barely beats seeding at random. That is a fact about this $c^\star$ and not a theorem about targeting. The spillover is $\delta_{\text{spill}}(c^\star) = 0.00792$.

Over the 400 sampled NROY points, the image of the policy effect is $\Theta_{\text{hm}} = [0.00577, 0.07369]$ and the image of the spillover is $\Theta_{\text{exp}} = [-0.000578, 0.026525]$. The report for θ_2 that the targets support is this range. It is neither a sampling distribution nor a posterior. Figure 2 shows every sampled point; for θ_2 the pointwise Monte Carlo error bars are smaller than the dots. A fresh recheck of the four endpoint candidates moves the ends of Θ_{hm} only in the sixth decimal and the ends of Θ_{exp} by amounts comparable to their Monte Carlo errors, and it changes no verdict (Appendix D).

Why is the image wide? Largely because of the assumption δ. With $\delta_1 = 0.02$ on a level moment, the rule lets each level miss by up to about $\sqrt{\tau\delta_1^2}$ on its own. That allowance exceeds $\theta_1(c^\star) = 0.03631$, the very effect the pilot level carries, so the pilot cannot pin the seeding effect κ. The spread ratio for κ, 0.962, is what that reading predicts, and the sensitivity menu of Section 7.5 confirms it by tightening δ_1. The peer weight is a separate case: a check at $c^\star$ found that the peer-correlation moment hardly moves across the box's range of γ; that check is not a committed result.

Table 5. One panel at each diagnostic point. The first column is the field experimenter; the other two are the calibrator. Each analyst row gives the interval, then containment of the truth at the point (a), containment of the truth at the village (b), and the verdict against the analyst's own image.

	$c^\star$ **(village)**	**min-I (**$I = 0.0141$**)**	**median-I (**$I = 3.3437$**)**
Policy effect θ_2*; image* $\Theta_{\text{hm}} = [0.00577, 0.07369]$			
truth $\theta_2(c_0)$	0.03826	0.03933	0.05642
A0 (Foil 1)	[0.03267, 0.03479]	[0.03057, 0.03285]	[0.04664, 0.04840]
(a) / (b) / verdict	no / no / F-OVERCONF	no / no / F-OVERCONF	no / no / F-OVERCONF
A1	[0.02636, 0.04938]	[0.02611, 0.05018]	[0.04287, 0.06781]
(a) / (b) / verdict	yes / yes / F-OVERCONF	yes / yes / F-OVERCONF	yes / no / F-OVERCONF
A2	[0.02518, 0.05071]	[0.02489, 0.05165]	[0.04171, 0.06880]
(a) / (b) / verdict	yes / yes / F-OVERCONF	yes / yes / F-OVERCONF	yes / no / F-OVERCONF
Spillover δ_{spill}*; image* $\Theta_{\text{exp}} = [-0.000578, 0.026525]$			
truth $\delta_{\text{spill}}(c_0)$	0.00792	0.01271	0.01520
Foil 2	[−0.001405, 0.017638]	[0.002988, 0.023619]	[0.005598, 0.026135]
(a) / (b) / verdict	yes / yes / no verdict	yes / yes / F-OVERCONF	yes / yes / F-OVERCONF

Intervals at nominal level 95% from the one panel at stream offset 0. The flags describe that one panel; they are not coverage. At $c^\star$ the verdict is reported for completeness and is not a test of the field experimenter's estimator. Foil 2 targets δ_{spill} and is compared only with Θ_{exp}.

7.3 One panel at each point

Table 5 reports every interval in the one panel drawn at each diagnostic point, with both containment flags and the geometric verdict. Figure 3 draws the two foils against their images.

The field experimenter, at $c^\star$. Her A0 interval is [0.03267, 0.03479] and the truth is 0.03826, so the interval misses; the point estimate is 11.9% below the truth. This failure needs no calibration. It is layer (a), and it comes from the identification conditions of the estimator: in this panel about 95% of the gap is the effect of the campaign on people who were not seeded, for which A0 has no term (Appendix B). Her A1 and A2 intervals, computed from a pilot panel at the same point, both contain $\theta_2(c^\star)$. Her Foil 2 interval contains the spillover $\delta_{\text{spill}}(c^\star)$. Against the images, the A0, A1 and A2 intervals all sit strictly inside Θ_{hm} and the Foil 2 interval starts below the lower end of Θ_{exp}, so it has no verdict. We report these verdicts and draw no charge from them: she worked with the village's own panel.

The calibrator, at min-I and median-I. At both points the A0 interval misses the truth at her own point (layer a) and the truth at the village (layer b), and it lies strictly inside Θ_{hm}. Her A0 interval is therefore wrong about her own world and also overconfident: it is far narrower than the range of effects that the targets leave open. The A1 and A2 intervals behave differently. At both points they contain the truth at her point. At the min-I point they also contain the truth at the village, because the two truths are close (0.03933 and 0.03826). At the median-I point they do not: the truth there, 0.05642, is far from the village's 0.03826, and the assumption "my point is the world" is costly. Even

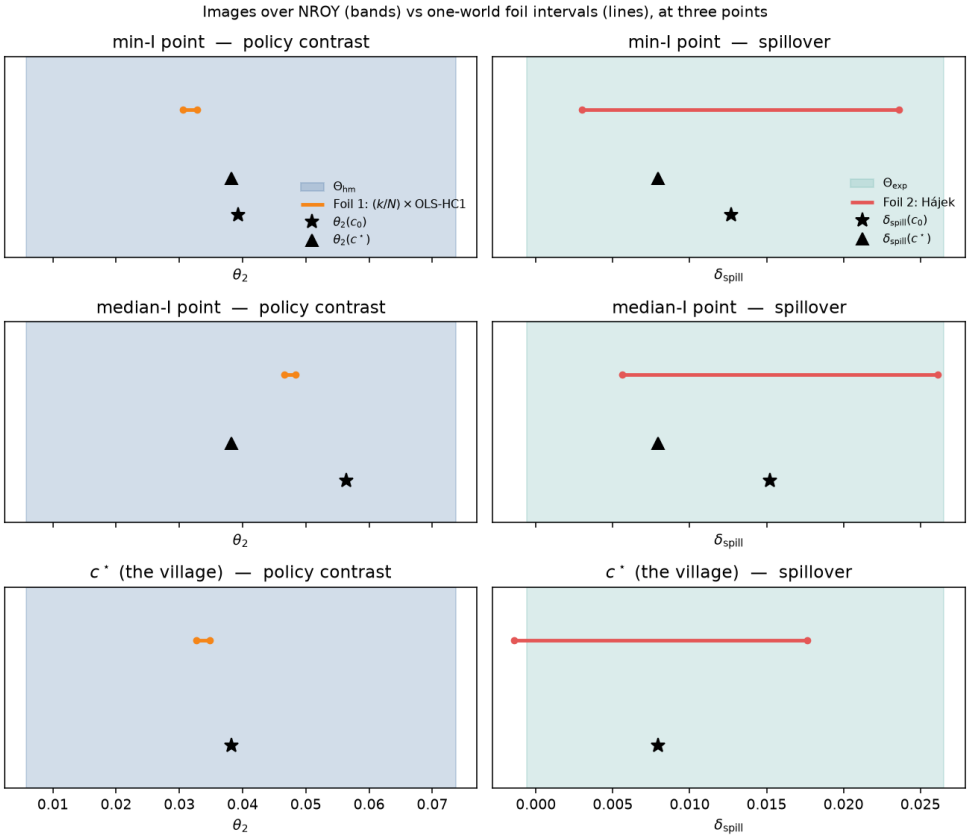


Figure 3. The two foils against their own images at the three diagnostic points (rows). Left: θ_2, the image Θ_{hm} (band) and the A0 (Foil 1) interval. Right: δ_{spill}, the image Θ_{exp} and the Foil 2 interval. A star marks the truth at the point and a triangle the truth at $c^\star$ (omitted in the $c^\star$ row). The two columns use different scales and different functionals; they are never compared with each other.

so, the A1 and A2 intervals are F-OVERCONF at both points. They are right about her world and still narrower than what the targets leave open. Foil 2 tells the same story for the spillover: its interval contains both truths at both points and is F-OVERCONF against Θ_{exp} at both.

Containment and geometry are different tests. The A1, A2 and Foil 2 rows show why the two checks must be kept apart. An interval answers a question about one world; the image answers a question about every world the targets cannot rule out. A one-world interval can be right about its world and narrower than the image at once. A calibrator who reports such an interval as her uncertainty about the village understates what the targets leave open. At the min-I point she happens to be right about the village as well; the median-I point shows that membership in NROY does not guarantee it, and the image is what warns her.

Table 6. The three analysts of θ_2 over 200 repeated panels at each point (amendment A2).

	Bias	SD	RMSE	Mean SE	SE/SD	Containment [CP 95%]	Width	Width/Θ_{hm}
min-I point, $\theta_2 = 0.03933$								
A0	−0.00793	0.00060	0.00796	0.00061	1.006	0.000 [0.000, 0.018]	0.00237	0.035
A1	−0.00168	0.00215	0.00272	0.00630	2.936	1.000 [0.982, 1.000]	0.02471	0.364
A2	−0.00130	0.00312	0.00337	0.00706	2.265	1.000 [0.982, 1.000]	0.02769	0.408
median-I point, $\theta_2 = 0.05642$								
A0	−0.00912	0.00048	0.00913	0.00048	0.994	0.000 [0.000, 0.018]	0.00186	0.027
A1	−0.00157	0.00210	0.00262	0.00652	3.107	1.000 [0.982, 1.000]	0.02556	0.376
A2	−0.00124	0.00278	0.00304	0.00713	2.564	1.000 [0.982, 1.000]	0.02797	0.412
$c^\star$ *(village),* $\theta_2 = 0.03826$								
A0	−0.00483	0.00057	0.00486	0.00057	1.000	0.000 [0.000, 0.018]	0.00223	0.033
A1	−0.00084	0.00203	0.00219	0.00603	2.973	1.000 [0.982, 1.000]	0.02363	0.348
A2	−0.00051	0.00294	0.00298	0.00674	2.291	1.000 [0.982, 1.000]	0.02642	0.389

Bias is the mean estimate minus $\theta_2(c_0)$. SD is the standard deviation of the estimates across panels (the sampling dispersion); Mean SE is the mean reported standard error; Width is the mean width of the reported interval. Containment is the share of panels whose interval contains $\theta_2(c_0)$, with an exact Clopper–Pearson 95% interval; for A1 and A2 it reflects conservative intervals, not calibration.

7.4 The price of allowing for interference

Table 6 reports the three analysts of θ_2 over $R = 200$ fresh panels at each point, and Figure 4 plots the absolute bias against the mean interval width. No panel left A2 undefined: the undefined count is 0 at the min-I point, 0 at the median-I point and 0 at $c^\star$.

Bias and spread. Absolute bias falls from A0 to A1 to A2 at every point, and the sampling SD rises from A0 to A1 to A2 at every point. A0's bias is large relative to its own spread: at $c^\star$ the bias is −0.00483 with a sampling SD of 0.00057, so its intervals almost never reach the truth. A1 and A2 are biased downward at every point, by much less. The amendment's proposed reason is two-hop spillover in the S_{0a} control: people with no seeded neighbor are still lifted by seeds two steps away, which raises the control mean; the independent review of the code checked the sign of the bias and not this mechanism. Allowing for interference does not remove the bias. It shrinks it, and the analyst pays with dispersion. On mean squared error the trade does not run all the way down the ladder: A1 has the smallest RMSE at every point, so A2's further cut in bias costs more in spread than it saves.

The reported intervals are conservative. A0's reported standard error matches its sampling SD closely (SE/SD of 1.006, 0.994 and 1.000). For A1 and A2 the reported SE is between 2.265 and 3.107 times the sampling SD. The A1 and A2 intervals are conservative relative to their sampling distributions. The split of the reported variance that follows is descriptive and is not a pre-registered result. It comes from addendum A2.1, which was designed after the A2 results and the independent review of the code were seen, states no expectation, and changes no estimator, interval or verdict (Appendix G). On the $c^\star$ panel, a share 0.886 of A1's reported variance, and a share 0.828 of A2's, is

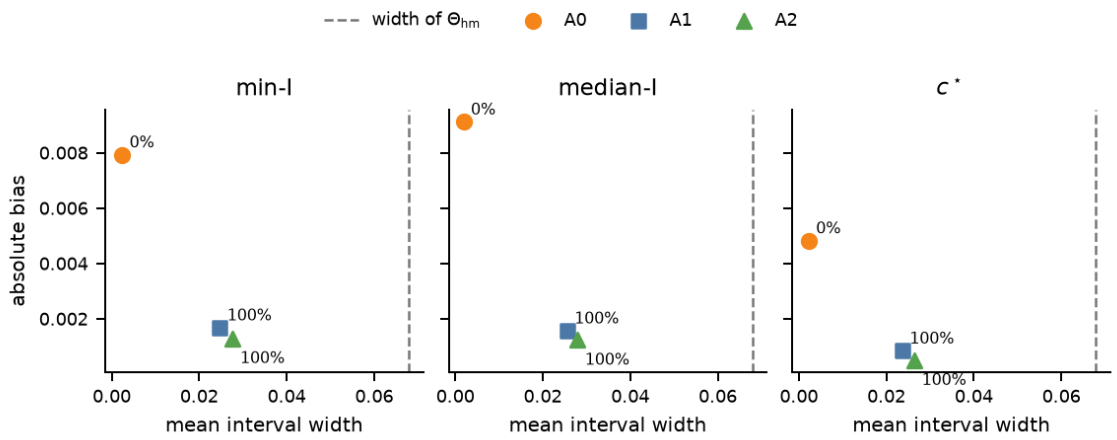


Figure 4. Absolute bias against mean interval width over 200 repeated panels, one marker per analyst, one panel per diagnostic point. The dashed line is the width of Θ_{hm}. The percentage beside each marker is the repeated-sampling rate at which the analyst's interval contains $\theta_2(c_0)$, printed as amendment A2 asked; for A1 and A2 it is conservative containment under the Young bound and should not be read as calibrated coverage.

the Young bound on pairs with a zero simulated joint exposure probability. The independent review of the code (`reviews/a2-code.md`) traced the largest piece to the S_{0b}–S_{0a} pair rather than the S_1–S_{0a} pair. The $\pi > 0$ term alone is smaller than the design variance of the A1 estimate (fixed noise, 400 assignment redraws, mean main-term SE^2 over design variance 0.801; full mean SE^2 over design variance 7.742). The same pattern holds for both analysts at all three points: the Young share runs from 0.816 to 0.896, the main-term ratio from 0.719 to 0.803, and the full ratio from 5.080 to 8.300. The Young terms are part of the pre-registered variance and are not retuned here; a sharper variance would be a new design, written before a new run.

Containment over repeated panels. A0's intervals contain $\theta_2(c_0)$ in none of the panels at any point. A1's and A2's contain it in every panel at every point, so the two are tied. That rate sits above the nominal 95% and reflects the conservative variance; it is not evidence that either interval is calibrated. Every repeated-sample mean width is narrower than Θ_{hm}, the widest at 0.412 of its width. That is interval geometry. It is not evidence that the interval is honest about what the targets leave open, and it is not a coverage claim.

A post hoc reading of the trade-off. What follows is an interpretation, formed after the A2 results and the A2.1 split were seen; it was not pre-registered and it tests nothing. Moving from A0 to A1 and A2 does three separate things. (i) It shrinks the bias but does not remove it. (ii) It raises the sampling SD. This is a real rise in dispersion, because the SD is measured across panels and does not depend on the variance estimator. (iii) It widens the reported interval by more than the rise in SD alone would, because of the Young bound. Under the pilot's design some pairs of people are never jointly exposed to the conditions that a contrast compares, so the design cannot identify the covariance of their outcomes, and the variance estimator replaces that covariance with Young's bound (Aronow and Samii, 2017). Part (iii) depends on the exposure map. Foil 2's spillover contrast uses the same pilot and the same simulated joint probabilities, yet on the $c^\star$ panel only a share 0.127 of its variance is the

Table 7. The five expectations of amendment A2, written before any A2 number, checked against Table 6.

	Expectation (at every point)	Finding
(a)	|bias| falls from A0 to A1 to A2	met at all three points
(b)	sampling SD and width rise from A0 to A1 to A2	met at all three points
(c)	A0 coverage far below nominal; A2 the closest to nominal	met only as a tie (A1 = A2)
(d)	A1 and A2 biased downward	met at all three points (sign only)
(e)	every interval narrower than Θ_{hm}	met at all three points

(b): SD is the sampling dispersion and width is the conservative interval; both rise, and they are different facts. (c): A0's rate is 0.000 at every point; A1 and A2 both reach 1.000, so A2 is the closest only jointly with A1, and both sit above nominal because their SE is between 2.265 and 3.107 times the SD. (d): the review checked the sign; the two-hop mechanism is the amendment's proposed reason. (e): mean width in Table 6; the one-panel intervals are also strictly inside Θ_{hm} (Table 5).

bound (Appendix C); A1 and A2 put weight on the S_{0b} stratum, where most of the bound's mass sits. The intervals that the bound widens are conservative, and the containment rate of one comes from the bound. They are not calibrated.

Table 7 collects the five expectations. Four are met as written at every point. Expectation (c) is met only as a tie: A0's rate is far below nominal, and A1 and A2 share the rate closest to nominal. The ladder therefore behaves as the amendment predicted in its first-order features, bias down and spread up, and it does not deliver a calibrated interval at any rung. More important for the paper's question, no rung produces an interval as wide as the image. Even the analyst who allows for interference and transports only within degree bins answers a question about one world.

7.5 How the image moves with the level tolerance

The image is a set under assumptions, and the assumption that appears to bind is the level tolerance δ_1. Amendment A1, written after the main results were known (Appendix G), reruns history matching and the image with δ_1 set to 0.02 (the main design), 0.01 and 0.005; Appendix E gives the tables. The kept set shrinks from 1038 points to 667 and 432, and at the smallest tolerance the image of the policy effect narrows to $[0.01573,\ 0.06186]$ while the spread ratio for κ falls to 0.606. That supports the reading that the level allowance is what leaves κ open. The verdicts at $c^\star$ do not move at any entry. A smaller tolerance is a stronger assumption, that the kernel matches the targets more closely; it buys a smaller set and asks the reader to believe more. The menu is reported, not used to choose δ, and the results of this paper stay at the main design.

SECTION 8

Relation to partial identification

The image borrows the reporting logic of partial identification, and it is worth being precise about how far the analogy goes.

A menu of answers indexed by assumptions. In the partial-identification tradition, an empirical question is answered by the set of values consistent with the data and a stated list of assumptions. Weaker assumptions give larger sets and more credible answers; stronger assumptions give smaller sets and less credible ones, the trade that Manski (2003) calls the law of decreasing credibility. Tamer (2010) describes the resulting practice as reporting a menu: one set for each set of assumptions a reader might accept, instead of one point under the strongest. This paper reports two such menus. The first is along the calibration axis. The image Θ_{hm} is the set of policy effects under the kernel, the box, the targets, τ and δ. Adding one more assumption, $c = c_0$, collapses it to the single value $\theta_2(c_0)$; that is what the calibrator does when she works at one point, and Table 5 shows the credibility cost at the median-I point. The sensitivity menu of Section 7.5 moves along the same axis by tightening δ. The second menu is along the interference axis. A0, A1 and A2 are three answers to the question of θ_2 from one design, indexed by how much they assume about the way outcomes respond to other people's seeding: no response at all, response through an exposure map with transport, and the same with transport restricted to degree bins. Weaker assumptions about interference buy less bias and cost precision, which is the same trade in a different currency.

Where the set comes from. In much of the literature the set comes from an incomplete model. A game with multiple equilibria does not say which outcome occurs, so the model is consistent with a set of outcome distributions, and the data bound the parameters without pinning them down (Tamer, 2003, 2010). Here the source is different. The kernel is complete: for each c and each draw of the noise it returns one panel, and the update map is a contraction with a unique stationary regime (Appendix A). The set arises because the targets do not restrict the parameters. Only α is pinned; the level tolerance exceeds the pilot's own effect, so κ stays open; and the peer-correlation moment does not restrict γ. Many complete worlds fit the same targets, and they disagree about the policy.

Where the analogy stops. The image is not an identified set. An identified set is a population object: the values of a parameter consistent with the distribution of the observables and the assumptions. The image here is a finite empirical range produced by thresholding simulated moment discrepancies for a fully specified kernel, evaluated on a sample of kept points. Its threshold mixes Monte Carlo variance with a design tolerance, so it does not correspond to a sampling distribution of the targets. A thresholded criterion resembles the level sets of Chernozhukov et al. (2007), and the range of θ_2 over such a set is a projection of the kind studied by Imbens and Manski (2004) and Kaido et al. (2019); see Molinari (2020) for a survey. Those papers give coverage guarantees under their assumptions. Nothing here claims one. Similarly, no analyst on the interference ladder reports an identified set for θ_2; each reports a point estimate with a standard error under its maintained assumptions. A bounds analysis of θ_2 from the pilot in the spirit of Manski (2013) would be a different and more conservative exercise. What the paper borrows is the rule for reporting: do not replace a set by a convenient point.

SECTION 9

What the procedure is not, and its limits

The procedure is easy to over-read, so we list what it does not claim.

It is not a posterior or a confidence set. NROY rules points out; it does not weight the points that remain, and Θ_{hm} is a range, not a distribution of effects. The threshold τ is the 0.95 quantile of a χ^2_4 distribution, but V is diagonal while the two level moments share a state stream, and the discrepancy δ rather than sampling noise fills most of V. So τ is a tolerance, not a test level, and the image carries no coverage guarantee. Approaches that do weight the parameter space, such as Bayesian calibration with an explicit discrepancy term (Kennedy and O'Hagan, 2001), answer a different question.

It is not a coverage claim for the one-panel intervals. The two flags describe one draw. The only coverage evidence is the Foil 2 calibration check (Appendix C) and the repeated panels of Section 7.4, where the A1 and A2 rates reflect conservative intervals rather than calibration.

It does not make A1 or A2 correct. A1 and A2 are biased downward at every point; transport shrinks the bias and does not remove it. Neither identifies θ_2 from the pilot, and their intervals target θ_2 under assumptions that the kernel does not satisfy exactly. They do not estimate the same functional as A0, which equals θ_2 only under no interference.

It is not a criticism of standard practice. A regression with robust standard errors quantifies uncertainty within one panel under the analyst's stated assumptions, and it does that well. An exposure-mapping estimator does the same under its own assumptions (Aronow and Samii, 2017; Leung, 2020), and recent work studies what such estimators recover when the map is wrong (Sävje et al., 2021) and how to test hypotheses about interference directly (Athey et al., 2018). The overlay asks a different question: whether one parameter point, or one world, is enough for a policy report. The failures we document come from applying a one-world tool to a question about the set of worlds the targets cannot rule out.

The analysts' panel overlaps the targets. The analysts' one panel is stream offset 0, which is also one of the 200 replications whose mean defines z. The analysts therefore see data that helped set the targets, with a weight of $1/200$. We did not rerun the pipeline without it, and we have not listed the rejected wave points just above τ, which a small shift in z could move into NROY.

It is one planted village. The project has one graph, one planted point and one set of targets, and the targets are the simulator's own output. It makes no claim about any real campaign and claims no external validity. With external targets, δ becomes the allowance for a model that is not the world and must be defended; the targets would also need a stated population and a model of how they transport to the village of interest.

Who might use it. The composition is useful to anyone who reports a policy number from a calibrated agent-based, structural or microsimulation model. The one-point Monte Carlo error bar is a statement

about one world. The image is a statement about the worlds the targets cannot rule out, and its width is measured rather than assumed. An empty NROY is also a usable result: it rejects the pair of kernel and targets for the declared design. Where interference matters and the data come from a randomized pilot, the ladder of Section 5 makes explicit what each step from the pilot to the policy assumes, and the planted setting makes its cost visible. A design-based answer that relies on fewer of these assumptions needs randomization at the level of the policy itself, for instance across villages (Hudgens and Halloran, 2008; Baird et al., 2018).

SECTION 10

Conclusion

A calibrated simulator's answer to a policy question is a function of the parameter point at which it is evaluated, and the targets that a researcher can observe rarely pin that point down. In a planted village where the truth is known, history matching against the status quo and one randomized pilot keeps a large set of parameter points, and the image of the policy effect over them is wide. The image, rather than a point inside it, is the report that the targets support.

Against that image we graded intervals computed from one panel, and we kept two checks apart. The analyst who assumes no interference misses the policy effect at the real village, and at the calibrator's best-fitting points she misses again and is overconfident. Analysts who allow for interference through an exposure map and transport from the pilot cut the bias, at the price of more spread and of intervals that are conservative relative to it; at the calibrator's points, their one-panel intervals contain the effect at her point and are still narrower than what the targets leave open. Containment and interval geometry answer different questions, and both are needed.

The broader lesson is a reporting rule. A policy number from a calibrated model should travel with the set of numbers that the same targets cannot rule out, and an estimate of a network policy from a randomized pilot should travel with the assumptions that carry it from the pilot to the policy. Both make the answer less sharp, and both make it easier to believe.

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APPENDIX A

Model and design details

The graph and the policy vector. The code draws $A_{ij} \sim \text{Bernoulli}(p)$ with $p = 4/N$ for every ordered pair, takes $A \vee A^\top$ and sets the diagonal to zero, so an edge appears with probability $1-(1-p)^2 \approx 2p$. The drawn graph has 83 triangles over 25,918 connected triples, a global clustering coefficient of 0.0096 against 0.0100 expected for a random graph of this density. The graph is close to tree-like: two people with a common friend are rarely friends themselves. In the policy vector $\mathbf{d}_2$, ties in degree go to the lower index: 35 of the 36 people with degree 12 are seeded, and everyone above that degree is seeded. Given the graph, $\mathbf{d}_2$ is deterministic.

The update map is a contraction. The logistic function has slope at most $1/4$, and the peer mean $(AY)_i/k_i$ is an average, so its sup norm is at most that of Y. For two states under the same assignment and the same shocks, one step of Equation (7) therefore shrinks their sup-norm distance by a factor of at most $(\beta+\gamma)/4 \le (0.60+0.80)/4 < 1$ over the whole box. For a fixed assignment the process has a unique stationary regime and forgets its starting state geometrically; after $T = 20$ steps the starting state matters by at most that factor to the power T, far less than the smallest Monte Carlo standard error in Table D.1. With a sum of neighbors' outcomes in place of the average, the peer slope would grow with degree and the bound would fail for parts of the box. The bound uses only the kernel, the box and the horizon, all fixed before any run.

Random streams. Every random stream is seeded. Rep r of every arm shares one state stream, the starting state and all shocks, and each arm has its own assignment stream, so contrasts between arms are paired. The image replications use offsets 0 to $200-1$; the analysts' one panel is offset 0; the endpoint recheck, the Foil 2 gate and the repeated panels of amendment A2 each use their own disjoint block (A2's starts at 2000000). The pipeline is deterministic: a rerun reproduces every digit.

History-matching waves. Figure A.1 shows the waves. The minimum implausibility in the three waves is 0.0274, 0.0874 and 0.0141, and the median is 51.01, 11.23 and 13.54, against $\tau = 9.4877$. Later waves concentrate on the kept region, and the threshold does not move. The min-I point is the kept point with the smallest I; the median-I point is the point at position $\lfloor |\text{NROY}|/2 \rfloor$ when NROY is sorted by I.

APPENDIX B

Foil 1 and the gap decomposition

The OLS coefficient. At $c^\star$ the A0 coefficient is $\hat{\beta} = 0.3373$ with interval $[0.3267, 0.3479]$; at the min-I point it is 0.3171 with $[0.3057, 0.3285]$, and at the median-I point 0.4752 with $[0.4664, 0.4840]$. The scale factor is $K/N = 0.1$. Five statements stay with every report of A0 (Foil 1):

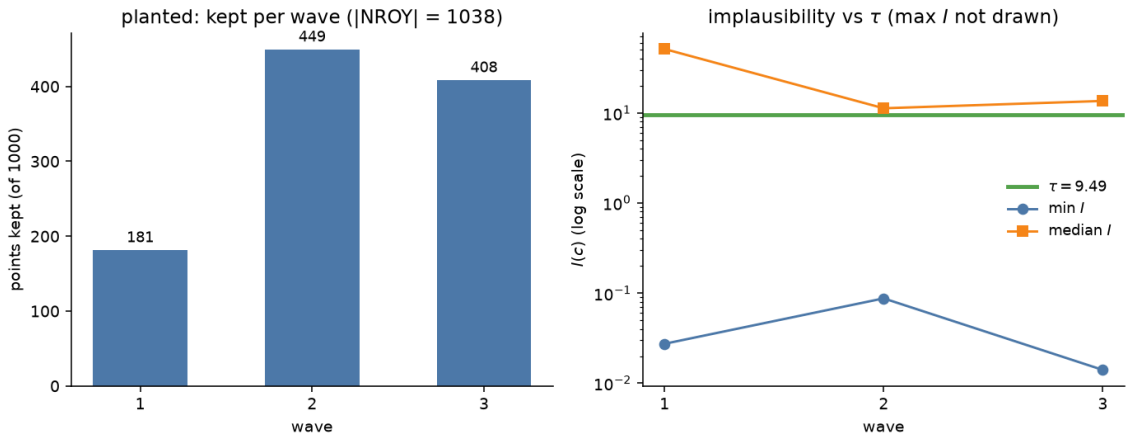


Figure A.1. History matching against the planted targets. Left: the points kept in each wave. Right: the minimum and median $I(c)$ in each wave against the threshold τ (log scale).

1. A0 is not an identified estimator of the simulator's θ_2 under the actual interference model. It is the whole-village policy estimate implied by an analyst who assumes no interference, $(K/N)\hat{\beta}$.
2. That reading assumes consistency and no interference, conditional exchangeability and positivity given X, a correctly specified additive constant-effect outcome model, and a sampling basis for HC1. The deterministic top-degree assignment does not establish those assumptions.
3. After scaling, the A0 intervals lie strictly inside Θ_{hm} and exclude the truth at their own point at all three diagnostic points. F-OVERCONF is an assumption-contingent interval-geometry finding and not a coverage claim for θ_2 under interference.
4. Across 200 paired panels, about 90% of the gap between θ_2 and (K/N) times the mean OLS coefficient is the policy effect on unseeded people. For the one displayed panel, the shares are about 95% at $c^\star$, 93% at min-I and 92% at median-I.
5. The rescaling by K/N was adopted after the frozen numbers were inspected. Its justification is the aggregation identity that holds under no interference. The post hoc design correction is disclosed (Appendix G).

The decomposition. With paired streams, the policy effect splits exactly into a seeded part and an unseeded part. Let H be the seeded set and $y_2 - y_0$ the effect on one person in one replication. Then

$$\theta_2 = \underbrace{\tfrac{1}{N}\textstyle\sum_{i\in H}(y_2 - y_0)}_{\text{seeded part}} + \underbrace{\tfrac{1}{N}\textstyle\sum_{i\notin H}(y_2 - y_0)}_{\text{unseeded part}}, \tag{B.1}$$

averaged over replications. The unseeded part is zero under no interference. The seeded policy contrast is the mean of $y_2 - y_0$ over H; under interference it is not the effect of own seeding alone. Table B.1 gives the numbers. The gap for the mean coefficient averages over 200 panels, whereas the gap for the displayed coefficient uses the one panel that A0 sees.

Table B.1. Gap decomposition for A0 (Foil 1), 200 paired replications.

	$c^\star$	min-I	median-I
θ_2	0.03826	0.03933	0.05642
seeded part	0.03396	0.03224	0.04824
unseeded part	0.00430	0.00709	0.00818
seeded policy contrast	0.3396	0.3224	0.4824
mean OLS coefficient	0.3350	0.3146	0.4735
gap for the mean coefficient	0.00477	0.00787	0.00907
unseeded share of that gap	0.9028	0.9015	0.9015
OLS-error part of that gap	0.00046	0.00078	0.00089
gap for the displayed coefficient	0.00454	0.00762	0.00890
unseeded share of that gap	0.9488	0.9308	0.9185
baseline gap (seeded minus unseeded, Y under control)	0.0007	0.0007	0.0008

Most of the gap is the effect on people who were not seeded, and an analyst who assumes no interference has no term for it.

Why there is no flexible-regression foil. On the policy arm, seeding is a deterministic function of degree and the index tie-break. Conditioning on degree leaves almost no overlap: no seeded person has degree below 12, no unseeded person has degree above it, and only one unseeded person has degree exactly 12. Omitting degree leaves the assignment variable unadjusted. A forest can differ numerically from OLS, but it keeps the same blind spot to interference and does not identify θ_2 from this panel.

APPENDIX C

Foil 2: the estimand and the calibration check

The Hájek target and the kernel truth. The Hájek mean in Equation (11) targets the inverse-probability exposure contrast. The kernel truth δ_{spill} in Equation (12) averages the unweighted realized contrasts. The exposure probabilities vary widely across people in the two unseeded strata, so the two functionals weight people differently. On this run the two are close enough that no containment flag depends on the difference: in the panels we examined, outcomes hardly track the inverse-probability weights. That comparison was made offline on paired panels and is not a committed result. It is a fact about this kernel, graph and parameter point; we did not run another kernel. The exposure map uses neighbors' seeding, whereas the kernel's peer term uses neighbors' outcomes and the effect travels more than one step, so the map is not exact for the kernel. Foil 2 cannot target θ_2 directly: the policy arm has no lottery, so given the graph each person's assignment probability is 0 or 1 and inverse-probability weights are not defined.

The pre-registered calibration check. The script `scripts/check_foil2_coverage.py` is a required check of Foil 2's interval. Its gate uses outcomes that obey the exposure map exactly, with a true spillover of 0.03, and it draws 10,000 pilot assignments. The gate rule was written before the run: the gate fails if the ratio of the mean estimated variance to the empirical variance, plus 1.645 bootstrap standard errors, is below one. The gate passes, with a mean estimate of 0.03006 and a ratio of 1.034 (bootstrap SE 0.0154). The result is:

> In a fixed-world repeated-assignment check whose outcomes obey the exposure map exactly, the nominal 95% Foil 2 Wald interval covered 93.4% of assignments (exact 95% binomial interval 92.9%–93.9%). The average estimated variance was 1.034 times the empirical design variance, so the shortfall is not explained by average variance underestimation. It reflects the finite-sample behavior of this studentized normal interval, with the implemented Hájek linearization and simulated joint exposure probabilities.

The check is specific to this implementation and is not a test of the theorem of Aronow and Samii (2017), whose exact result is for Horvitz–Thompson totals with exact joint probabilities. The code uses Hájek means, a first-order linearization and one simulated table of joint probabilities.

The script also reports the kernel itself, where the exposure map is not exact. Over 200 kernel panels, the nominal 95% interval contains the truth at a rate of 0.920 at $c^\star$, 0.925 at the min-I point and 0.920 at the median-I point. Redrawing only the assignment while holding the noise fixed gives 0.930, 0.9267 and 0.9267. The code reports these numbers and does not gate on them.

The spillover contrast is the case in which the Young bound matters least. In the descriptive variance split of addendum A2.1, designed after the A2 results, the Young share of the spillover contrast's variance on the $c^\star$ panel was small, 0.127, and its mean full SE^2 over the fixed-noise design variance was 1.048, whereas the A1 and A2 contrasts include the S_{0b} stratum and carry a much larger Young share (Section 7.4). This is consistent with Foil 2's interval covering a little under nominal in its check while the A1 and A2 intervals are conservative.

APPENDIX D

The endpoint recheck

Each image endpoint is an extremum of 400 noisy estimates, so the standard error attached to the extremizing point is not a standard error for the endpoint: it ignores the selection of the extremum. The code evaluates each endpoint candidate again with 200 fresh replications on separate streams (Table D.1).

Fresh re-evaluation removes the reuse-of-noise bias for the selected candidates. It is not simultaneous inference for the true extrema, and it says nothing about extrema that the sample missed. Does the endpoint noise change any verdict? For Θ_{hm} the fresh values differ from the original ones only in the sixth decimal. For Θ_{exp}, put the fresh values in place of the original endpoints: the lower end of each calibrated Foil 2 interval is still above the lower endpoint, the upper end is still below the upper endpoint, and the $c^\star$ interval still starts below the lower endpoint. None of the classifications changes. This is a comparison of printed numbers, not a field of the results.

Table D.1. Independent recheck of the endpoint candidates. SE is the pointwise Monte Carlo standard error.

Endpoint	NROY index	Original (SE)	Fresh (SE)
Θ_{hm} lower	293	+0.005771 (0.000008)	+0.005772 (0.000007)
Θ_{hm} upper	832	+0.073692 (0.000006)	+0.073697 (0.000006)
Θ_{exp} lower	987	−0.000578 (0.000505)	+0.000200 (0.000494)
Θ_{exp} upper	674	+0.026525 (0.000613)	+0.027542 (0.000601)

APPENDIX E

The sensitivity menu for the level tolerance

This analysis was decided after the main results were known. Amendment A1 of the pre-registration was written once the main run had reported Θ_{hm} and the spread ratios, and once the allowance $\sqrt{\tau\delta_1^2}$ had been set beside $\theta_1(c^\star)$. It tests no hypothesis and is never used to choose δ.

The menu sets the discrepancy of the two level moments to 0.02 (Main), 0.01 (Half) or 0.005 (Quarter). The two correlation moments keep $\delta = 0.05$. For each entry the code reruns the waves with the same seed, NROY, the 400-point image sample and the spread ratios; τ, C, the targets and every seed are unchanged. The Main entry reproduces the main run exactly, and the script stops with an error if it does not. The analysts are not rerun at the calibrator's points, because those points move with δ; the field experimenter's intervals at $c^\star$ do not depend on δ, so they are only classified against each entry's images.

Table E.1. The image under three level tolerances (amendment A1).

Entry	δ_1	|NROY|	Θ_{hm}	Θ_{exp}
Main	0.02	1038	[0.00577, 0.07369]	[−0.000578, 0.026525]
Half	0.01	667	[0.00637, 0.07432]	[−0.000576, 0.027605]
Quarter	0.005	432	[0.01573, 0.06186]	[0.000110, 0.021756]

Table E.2. Spread ratios, the history-matching finding, and the field experimenter's verdicts at $c^\star$ under the same entries.

Entry	α	κ	γ	Finding	A0 at $c^\star$	Foil 2 at $c^\star$
Main	0.259	0.962	0.992	none	F-OVERCONF	no verdict
Half	0.169	0.904	1.005	none	F-OVERCONF	no verdict
Quarter	0.111	0.606	0.952	none	F-OVERCONF	no verdict

Finding is `finding_hm`: "none" means neither F-EMPTY nor F-SINGLETON. The verdicts at $c^\star$ are reported for completeness and are not a test of the field experimenter's estimators.

The tolerance matters most for κ and α. The kept set shrinks as δ_1 falls, and at Quarter both images are narrower than at Main. The spread ratio for κ falls to 0.606 and the one for α to 0.111, which fits the reading that the level allowance is what leaves κ open. The ratio for γ stays near one

(0.952 at Quarter), but the menu leaves the correlation tolerance alone, so it says nothing new about γ. The entries are not nested: the upper end of both images at Half sits above the one at Main, because each image is the range over a sample from a different finite NROY set, with the waves redrawn. We do not read small moves of an endpoint as a trend. The clear change is at Quarter, where the lower end of Θ_{exp} is above zero (0.000110); under Main and Half the sign of the spillover is open. No entry returns F-EMPTY or F-SINGLETON. The verdicts at $c^\star$ do not move, and a smaller δ_1 does not give Foil 2 a verdict at $c^\star$ even at Quarter.

This menu is the reporting rule of Tamer (2010) applied to δ: a set under each of several assumptions, instead of one answer. In this planted exercise the targets are the simulator's own output, so the smaller tolerances are not an unfounded claim here. With external targets they would have to be defended. We adopt none of the entries; the paper's results stay at Main.

APPENDIX F

Replication

Everything in the paper reruns with one command from the repository root, after a one-time setup of the pinned environment described in `README.md`:

```
.venv/bin/python scripts/replicate.py
```

The script runs the test suite, checks that the planted targets regenerate byte for byte, runs the planted pipeline, the Foil 2 calibration check, the sensitivity menu (amendment A1), the interference trade-off (amendment A2) and its descriptive variance split (addendum A2.1), redraws the figures from the saved results, copies them into `writeup/figures/` and `thesis/figures/`, writes both copies of `numbers.tex`, builds the earlier write-up, and compares every result file with the committed version. Every random stream is seeded, so a rerun reproduces every digit; a mismatch means that the code or the environment changed. The option `--quick` skips the Foil 2 check, both amendments and the addendum. This paper is then built from `thesis/` with `latexmk`. The measured runtimes and package pins are in `README.md`. The exogenous design lives in `config/world.toml` and `config/design.toml`; the targets in `data/planted.json`; the results in `results/`.

APPENDIX G

Disclosures

The pre-registration and its amendments. The design is `docs/preregistration-v2.md`, frozen before any run of this version of the project. Two amendments and one addendum were written later, and each says what was already known when it was written. Amendment A1, the sensitivity menu of Appendix E, was written after the main results were known; it tests no hypothesis. Amendment A2, the analysts A1 and A2 and the repeated-panel exercise of Section 7.4, was written after the main and A1 results were known, including A0's gap decomposition and Foil 2's calibration check,

and before any A2 code or number existed. Its five expectations were written then and are reported as found. A later erratum to A2 changed wording only: the pilot realizes the policy vector with a positive and negligible probability, where the first wording had said that it never does. Addendum A2.1 was written after the A2 results and after the independent review of the A2 code, which had found the Young bound with a probe at one point. It turns that probe into a committed, descriptive split of the reported variance, with a fixed-noise redraw of the assignment (Section 7.4). It changes no A2 estimator, interval, flag or verdict, and it states no expectation.

Post hoc choices in the main design. The rescaling of A0's coefficient by K/N was adopted after the frozen numbers were inspected; its justification is the aggregation identity under no interference. An earlier version of Foil 2 used a first-order variance that dropped the covariance between strata; commit `36ed3e6` replaced it with the Aronow–Samii variance, and no coverage number from the earlier version is cited. The analysts' panel is also one of the replications that define the targets (Section 9).

Independent checks. The project's rule is that every statistical claim is checked by a reviewer running on a model family different from the one that produced it. The reviews are in `reviews/`. The wording of the A1 and A2 results in Section 7.4 follows the required wording of the review of amendment A2 (`reviews/a2-code.md`).

Use of AI assistance (DRAFT for the author to edit). *[Draft.]* The author designed the project, chose the questions, the model, the targets and the evaluation, and approved every design decision and amendment. AI assistants were used extensively under the author's direction: to write and test the simulation code, to run the pipeline, to draft and edit the text of this paper, and to review code, statistics and text. The producing and reviewing roles were assigned to models from different families (Anthropic Claude, OpenAI models through Codex, xAI Grok, Google Gemini and DeepSeek), and each review is committed in the repository. Every number in the paper is generated from the committed results by a script, and the full pipeline replicates with one command. *[The author to state here what he reviewed personally, and to confirm the statement against the Arizona State University guidance on AI use before the paper is circulated.]*